

Aditya Rao

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EDUCATION

McMaster University

B.Eng in Software Engineering (CO-OP)

Hamilton, ON

Sept 2023 – Present

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math I & II, Intro to Software Design, DSA, Computer Architecture, Software Development Basics

EXPERIENCE

Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Developing WebSphere Application Server v8.5.5, v9.0, WebSphere Liberty, and OpenLiberty and the public and internal tools which support development.
- Developed an upload feature in our translation tool which allows developers to upload translations for different languages and regions.
- Modernized the translation tool to leverage the latest technologies on OpenLiberty using Jakarta EE CDI, Restful, Concurrency, and MicroProfile Config, Telemetry, containerizing the application, and added Unit and Integration tests to ensure correctness of the workflow
- Added Apache to the translation tool as the reverse proxy (for both frontend and backend) to support TLS certificate management and OAuth authentication replacing the existing LDAP Basic Authentication
- Optimized the core data processing tool within this suite, replacing heavy framework overhead with custom API wrappers to reduce execution time by 99% (from 10 mins to <4 secs).
- Re-engineered the other 3 legacy internal tools from scratch, modernizing UX, using the latest Spring Boot and Jakarta EE technologies as above, standardizing Docker deployments, and establishing the first Unit/Integration testing pipelines to eliminate manual verification.
- Streamlined the installation FAT suite for WebSphere Application Server by removing complex infrastructure dependencies, significantly reducing debugging overhead and simplifying long-term maintenance.
- Diagnosed and solved >50 build breaks and supported the development team in diagnosing customer cases and build breaks on all platforms, including but not limited to Linux, Windows, MacOS, IBM i, and zOS
- Solved a customer case by replicating a customers environment by created E2E tests for deploying Liberty on OpenShift and verifying behavior such as Infinispan and Keycloak.
- Created a tool used by over 400+ developers and managers which allows developers to record the distribution of their work they perform and allow managers to record it and use the data to assist the developers by allocating resources in their workflows.
- Created an AI Agent leveraging IBM Bob Shell to analyze build breaks given the build break issue and description, build break logs and server configuration, OpenLiberty and WebSphere Liberty source code reducing time spent on fixing build breaks by at least 2 days worth of analysis.
- Completed a feature which allows Modernized Runtime Extensions for Java to add custom configuration actions to their WebSphere Application Server Network Deployment instances to run on server startup for Profile Conformance
- Completed a feature which allows developers to install versionless features using installUtility on WebSphere Liberty
- Cleaned up stale install issues on the OpenLiberty Github and WebSphere Liberty
- Offloaded, trained and transferred the translation tool to the build team in Kochi for easier maintainence (since they're the Client Zero for this tool)
- Generated knowledge transfer documents and trained the incoming intern to take over responsibilities and continue maintainence on these internal tools

Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Developed and maintained React and TypeScript code for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.
- Implemented real-time speech transcription using Python, OpenAI Whisper, FFmpeg, and Socket.IO

PROJECTS

WebCraft  <i>C++, CMake, Vcpkg, Asynchronous Programming, Github Actions</i>	Apr 2025 – Sept 2025
- Developed an asynchronous runtime and programming library using C++ 23 features to allow developers who are used to Node.js, Python, and Java to write fast and simple async C++ code easily. - Created coroutine abstractions, the async runtime, async streams, file and networking module abstractions over low level OS specific async library code to simplify cross-platform C++ development. - Created CI/CD pipelines and created over 130+ tests to verify correctness using gtest, CTest, Vcpkg, and Github Actions to ensure code quality and reliability. - Created a vcpkg port so developers can easily access it.	
Resume Builder  <i>Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker</i>	Aug 2024 – Feb 2025
- Developed an AI-powered resume builder for automated resume creation, scoring, and real-time editing. - Improved resume management by reducing manual effort in tailoring resumes for job applications.	
FashioNova   <i>Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask</i>	Oct 2024 – Oct 2024
- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough - Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed - Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.	
Smart Study Buddy   <i>Python, Flask, Wolfram Alpha API, OpenAI API, Next.js, Typescript</i>	Oct 2024 – Oct 2024
- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough - Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed - Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.	
QuakeGuard   <i>Next.js, Flask, Typescript, Python, TensorFlow, Keras, SciKit Learn, Pandas</i>	Jan 2024 – Jan 2024
- Developed an earthquake prediction tool using ML models showcased at DeltaHacks X and GDSC McMaster - Visualized global earthquake data and high-risk areas with Next.js and Python, Flask, SciKit Learn, and TensorFlow	

EXTRA CURRICULARS

Sys-Admin and Fullstack Developer Executive	Nov 2024 – Present
<i>IEEE McMaster Student Branch</i>	<i>Hamilton, ON</i>
- Set up an RPi 3B+ Cluster for a server to handle up to 100 clients in an instant with a budget of \$5000 - Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster - Developing the website for the IEEE student branch and the admin dashboard for executives to manage	

TECHNICAL SKILLS

Languages: Java, Python, SQL, .NET Framework, C, C++, JavaScript/TypeScript, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP
Developer Tools: VS Code, IBM Bob, Eclipse, AWS, GCP, Git, Github, Docker, OpenShift, Kubernetes, Databases, Infinispan
Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, Mongoose
Hobbies: Sketching, Painting, Violin (Western and Carnatic), Badminton, Swimming, Chess

CERTIFICATIONS

Clean Code	September 2025
<i>Udemy</i>	
IBM Agile Explorer	September 2025
<i>IBM</i>	
The DevOps Essentials - The Handbook	September 2025
<i>Udemy</i>	
Introduction to Containers, Kubernetes, and OpenShift	August 2025
<i>IBM</i>	
Practical OpenShift for Developers - OpenShift 4	August 2025
<i>Udemy</i>	
[NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025	February 2025
<i>Udemy</i>	
Docker & Kubernetes: The Practical Guide [2025 Edition]	January 2025
<i>Udemy</i>	